Gauge: Distribution-Free Conformal Coverage for IVIM Parameter Maps, and the Identifiability Wall in the Pseudo-Diffusion Compartment

The Gauge Project

Abstract

Intravoxel incoherent motion (IVIM) imaging fits a bi-exponential signal model to recover tissue diffusion D, pseudo-diffusion D^*, and perfusion fraction f. The pseudo-diffusion compartment is ill-posed, and model-based uncertainty quantification (UQ) for IVIM is documented to be overconfident in D^*. We ask whether *distribution-free conformal prediction* can supply the finite-sample coverage guarantee that model-based UQ lacks, and whether its advantage concentrates in the unstable compartment. On a controlled synthetic cohort built from our own forward model, we find that (i) model-based IVIM UQ is overconfident *broadly* across D, D^*, and f – not specifically in pseudo-diffusion – so the naive marginal hypothesis is *rejected*; (ii) conformal and conformalized-model-based intervals restore near-nominal marginal coverage in every compartment ($|\text{gap}| \leq 0.024$), and conformalizing a Casali-style mixture-density deep ensemble (MDN) is the sharpest valid recipe (0.65–0.79× the width of pure conformalized quantile regression at equal coverage); and (iii) the real failure is *conditional*: the high-D^* regime under-covers for every method at every SNR, and *no label-free method closes the gap.* A Fisher-information analysis localizes this to an **irreducible identifiability limit** – in the high-D^* regime the Cramér–Rao bound on D^* reaches the width of the regime bin, so no observable can route voxels to a wider-interval stratum. We further stress the guarantee under deliberate exchangeability breaks (weighted conformal recovers covariate shift but not forward-model misspecification; an observable deployment monitor fires before coverage silently fails, yet is blind to the latent high-D^* gap), show that the wall is robust to the b-value scheme, and give a qualitative in-vivo demonstration in which the same monitor turns the absence of ground truth into an observable stop sign. We frame the contribution as *characterizing* the high-D^* limit, not solving it.

1 Introduction

IVIM imaging [Le Bihan et al., 1988] separates microcirculatory perfusion from tissue diffusion by fitting the bi-exponential signal

$$\frac{S(b)}{S_0} = f\, e^{-bD^*} + (1-f)\, e^{-bD}, \tag{1}$$

where b is the diffusion weighting (s/mm^2), D the tissue diffusion coefficient, $D^* \gg D$ the pseudo-diffusion coefficient, and f the perfusion fraction. Equation (1) is notoriously ill-posed: D^* is poorly constrained because the fast-decaying perfusion term contributes only at low b, and small perfusion fractions make it nearly unidentifiable. Clinical use of IVIM parameter maps therefore demands trustworthy *per-voxel uncertainty*, not just point estimates.

The dominant approach to IVIM UQ is *model-based*: probabilistic neural networks, deep ensembles, mixture-density networks, and Bayesian fitting place a distribution over parameters and read intervals from it. These methods make distributional assumptions and carry *no finite-sample coverage guarantee*; the most comprehensive IVIM-UQ study to date [Casali, 2026] documents that such models are well-calibrated for D and f but *overconfident in* D^*. *Conformal prediction* [Vovk et al., 2005, Lei et al., 2018, Angelopoulos and Bates, 2023] offers the complementary property: under exchangeability of calibration and test data, it produces intervals with finite-sample, distribution-free marginal coverage for *any* base predictor – a biased base only widens the interval, it does not break coverage. Conformalized quantile regression (CQR) [Romano et al., 2019] makes these intervals input-adaptive. Conformal UQ has been applied to quantitative MRI relaxometry and field mapping [Birk et al., 2024], but never to the ill-posed IVIM inverse problem, and never benchmarked head-to-head against model-based IVIM UQ.

This paper closes that gap and, in doing so, *reframes* the question. We began with the hypothesis that conformal prediction's advantage would concentrate in the unstable D^* (and secondarily f) compartment, exactly where model-based UQ is documented weak. The data reject the *marginal* form of that hypothesis and replace it with a sharper, IVIM-specific finding. Our contributions are:

1. **Conformal/CQR coverage for IVIM** with validated finite-sample marginal coverage for (D, D^*, f) (Sec. 2).
2. **A conformal-vs-model-based benchmark** showing model-based IVIM UQ is overconfident *broadly*; conformal restores coverage in every compartment; and conformalizing the MDN is the sharpest valid recipe (Sec. 3.1).
3. **The conditional finding and its diagnosis**: the high-D^* regime under-covers universally, no label-free method closes the gap, and a Cramér–Rao analysis shows this is an *irreducible identifiability limit* (Secs. 3.2–3.3).
4. **Robustness, acquisition-sensitivity, and an in-vivo demonstration**: weighted conformal recovers covariate shift but not misspecification, an observable monitor flags deployment-validity breaks before coverage fails, the wall is robust to the b-value scheme, and the in-vivo demo is honestly qualitative (Secs. 3.4–3.6).

Every number in this paper traces to a gated, seeded checkpoint output; a machine-checked consistency report is included (Sec. 6).

2 Methods

Forward model and synthetic cohort. Conformal calibration requires *labeled* data, which in-vivo IVIM cannot provide, so the cohort is synthetic and built from our own implementation of Eq. (1) with Rician noise ($\sigma = S_0/\mathrm{SNR}$). Parameters are drawn uniformly from published abdominal ranges: $D \in [0.5, 3.0] \times 10^{-3}$, $D^* \in [10, 100] \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$, $f \in [0.05, 0.40]$; SNR is drawn from $\{10, 20, 30, 50, 100\}$; the b-value scheme has 22 values $(0, 10, \ldots, 100, 120, \ldots, 200, 300, \ldots, 800\,\mathrm{s/mm}^2)$, dense at low b to separate D^*. The three splits (train/calibration/test = 4000/2000/3000) are i.i.d. draws from one seed (`20260613`), making calibration and test exchangeable. On clean (noise-free) signals the generator and the non-linear least-squares (NLLS) estimator recover the truth to a maximum relative error of 2.07×10^{-7}, confirming label self-consistency.

Conformal intervals. For a base point predictor $\hat{\theta}$, *split conformal* uses calibration nonconformity scores $s_i = |\theta_i - \hat{\theta}_i|$ and the radius $q = \lceil (n+1)(1-\alpha) \rceil$-th smallest score, giving $[\hat{\theta} - q,\ \hat{\theta} + q]$

with marginal coverage in $[1-\alpha, 1-\alpha+1/(n+1)]$. *CQR* [Romano et al., 2019] conformalizes a gradient-boosted quantile-regression base. We also *conformalize model-based* bases by treating their predictive quantiles as the CQR band, which restores the guarantee while keeping the model's sharpness.

Model-based baselines. Four standard, independent model-based UQ baselines emit per-parameter intervals on the matched cohort: a single probabilistic NN (Gaussian NLL head); a deep ensemble of mixture-density networks [Bishop, 1994, Lakshminarayanan et al., 2017] following the Casali approach [Casali, 2026] with an aleatoric/epistemic split; a plain deep ensemble of point regressors (epistemic-only; deliberately weak); and a per-voxel Bayesian fit by component-wise adaptive Metropolis (fed the true noise level, so it is evaluated at its best). A parallel line of work uses normalizing-flow / simulation-based posterior estimation for microstructure inference; Gauge is deliberately *independent* of it (own forward model, own cohort, standard baselines) and does not use it as a baseline.

Conditional coverage and the identifiability diagnostic. We measure coverage on a (true-D^* tercile $\times$ SNR) grid – the terciles are a diagnostic only; the true D^* is unknown at test time. To ask whether a residual gap is a method failure or an information limit, we compute the Fisher information for $\theta = (D, D^*, f, S_0)$ under the high-SNR Gaussian approximation, $\mathcal{I} = J^\top J/\sigma^2$ with J the signal Jacobian, and the Cramér–Rao lower bound (CRLB) [Cramér, 1946] $\mathrm{CRLB}(\theta_k) = \sqrt{[\mathcal{I}^{-1}]_{kk}}$.

Robustness tooling. To stress exchangeability we use *weighted / nonexchangeable conformal* [Tibshirani et al., 2019, Barber et al., 2023], with importance weights $w(x) = \mathrm{d}P_{\mathrm{test}}/\mathrm{d}P_{\mathrm{cal}}$ estimated by a domain classifier on observable features, and *localized* [Guan, 2023] and *conditional* [Gibbs et al., 2023] conformal for the label-free conditioning attack. We also build a label-free *deployment-validity monitor* with two detector families (a Mahalanobis distance on observable signal-summary features and a residual-conformal test on NLLS fit residuals) that fires when an aggregate drift score exceeds a calibration-derived null threshold.

3 Results

3.1 Marginal benchmark: model-based UQ is broadly overconfident

Plain split conformal and CQR track nominal coverage within 0.03 for all parameters and all $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ on the held-out test split (e.g. split-conformal D^* at α=0.10: realized 0.897 vs nominal 0.90), confirming the conformal layer is correct. Table 1 gives the head-to-head at $\alpha = 0.10$.

Averaging the gap over α, the marginal under-coverage is broad, not D^*/f-concentrated: per-method worst compartments are f (PNN, 0.112), D (MDN, 0.048; DeepEnsemble-Point, 0.449), and f (Bayesian, 0.045). 0/4 **methods** show D^*/f-concentrated under-coverage; for the NN ensembles D^* is in fact among the *better*-covered parameters marginally, because its large aleatoric variance widens its band. The naive marginal hypothesis is therefore *rejected.* Conformal and conformalized variants restore near-nominal coverage everywhere, with maximum conformalized gap 0.024 (Fig. 1). On sharpness (Fig. 2), the cost of restoring coverage is cheap for strong bases (MDN 1.08–1.12×, Bayesian 1.02–1.05× the raw width) and ruinous for the overconfident weak base (DeepEnsemble-Point 3.17–3.72×). Crucially, the model's band helps: *conformalized-MDN is* 0.73/0.79/0.65× *the width of pure CQR* for $D/D^*/f$ at equal guaranteed coverage – the sharpest valid recipe.

Table 1: Marginal coverage on test at $\alpha = 0.10$ (nominal 0.90); gap = nominal − realized (positive = overconfident). Model-based methods under-cover broadly; conformal and conformalized variants restore coverage in every compartment. From `results/benchmark_report.txt`.

arm	D cov (gap)	D^* cov (gap)	f cov (gap)
raw: PNN-Gaussian	0.789 (+0.111)	0.823 (+0.077)	0.785 (+0.115)
raw: MDN-DeepEnsemble	0.856 (+0.044)	0.868 (+0.032)	0.853 (+0.047)
raw: DeepEnsemble-Point	0.431 (+0.469)	0.432 (+0.468)	0.483 (+0.417)
raw: Bayesian-MCMC	0.876 (+0.024)	0.870 (+0.030)	0.855 (+0.045)
conformal: split-NLLS	0.899 (+0.001)	0.897 (+0.003)	0.907 (−0.007)
conformal: CQR-HGB	0.901 (−0.001)	0.902 (−0.002)	0.893 (+0.007)
conformalized: MDN	0.891 (+0.009)	0.908 (−0.008)	0.892 (+0.008)
conformalized: Bayesian	0.895 (+0.005)	0.894 (+0.006)	0.876 (+0.024)

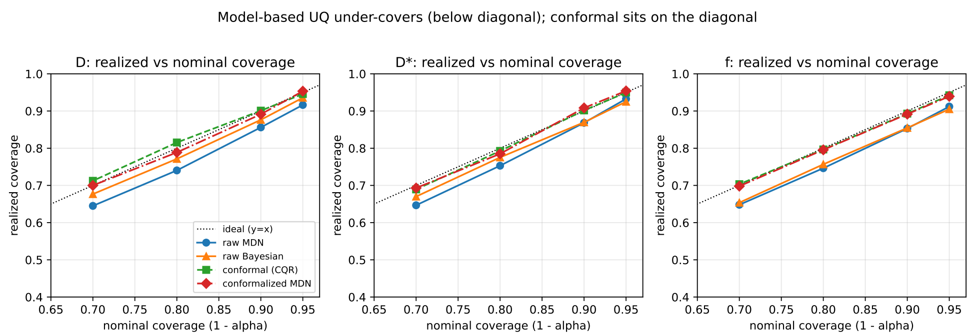

Figure 1: Realized vs nominal coverage per parameter. Raw model-based methods sit below the diagonal (overconfident); conformal and conformalized methods sit on it.

3.2 The conditional finding: high-D^* under-covers universally

Marginal coverage hides a compartment-specific failure (Fig. 3). On the (SNR × D^*-tercile) grid, the high-D^* tercile under-covers for *every* method at *every* SNR, while the mid-D^* tercile over-covers ($\approx$ 0.99). Table 2 gives the high-D^* slice: even the best method (conformalized-MDN) reaches only 0.877 marginal-in-regime and 0.808 in its worst-SNR cell. Critically, *Mondrian stratification by the known SNR does not fix it* – the failure axis is the *unknown* true D^*, not SNR. This is the refined, IVIM-specific form of the original hypothesis.

3.3 The high-D^* gap is an irreducible identifiability limit

We attacked the high-D^* gap with eleven *label-free* conditioning methods – plug-in Mondrian by estimated $\hat{D}^*$, localized conformal on observable features [Guan, 2023], and conditional conformal over an observable basis [Gibbs et al., 2023] – none of which uses the true D^* (Fig. 4). *None materially closes the gap.* The best label-free method (MDN with localized conformal) reaches 0.868 marginal / 0.819 worst-SNR, within sampling noise of the 0.808 baseline. The reason is structural, not methodological:

- **Routing error.** A plug-in $\hat{D}^*$ misroutes 31% of true high-D^* voxels to a lower-D^* (smaller-radius) stratum: the estimate is least reliable exactly where it must route correctly. Mondrian-by-$\hat{D}^*$ attains nominal coverage *conditional on the observed stratum* ($[0.90, 0.89, 0.90]$) yet craters on the *true* high-D^* tercile (0.76) – it controls the observable, not the latent axis.

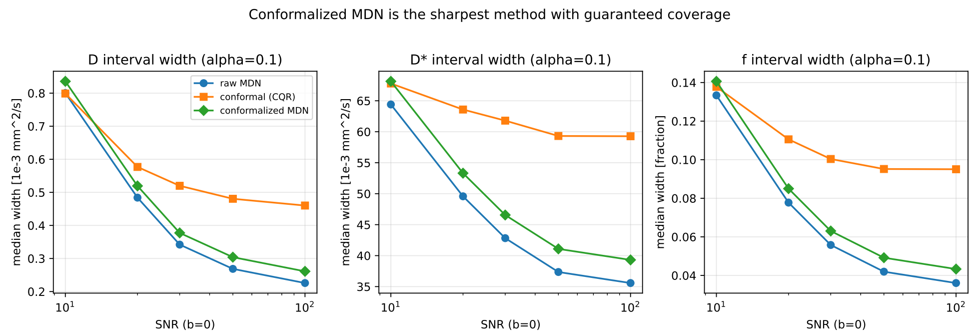

Figure 2: Median interval width vs SNR. Conformalized-MDN is the sharpest method with a guaranteed coverage level.

Table 2: High-D^* tercile coverage (D^* parameter, nominal 0.90). Every method under-covers; SNR-Mondrian does not close it. From `results/conditional_report.txt`.

arm	hi-D^* marginal	hi-D^* worst-SNR cell
raw-MDN	0.825	0.756
CQR (plain)	0.819	0.767
split (Mondrian/SNR)	0.808	0.764
CQR (Mondrian/SNR)	0.814	0.766
conformalized-MDN	**0.877**	**0.808**

- **The CRLB wall.** The CRLB on D^* grows $\sim 6\times$ from low to high D^*; the ratio of median CRLB(D^*) to the tercile bin width rises $0.34 \to 0.69 \to 1.12$ across terciles (Fig. 5). In the high-D^* regime the CRLB *exceeds* the bin width: a voxel's D^* cannot be localized to its own tercile from the data, so no observable can route it to a wider-interval stratum. Conformal interval width correctly tracks the per-voxel CRLB (log–log $r = 0.75$): the intervals are honest, not broken.

Verdict. High-D^* regime-conditional coverage in IVIM is an *irreducible identifiability limit*. We therefore *characterize* the limit; we do not claim to solve it.

3.4 Robustness under deliberate exchangeability breaks

We deploy one conformal predictor and one monitor, calibrated on the i.i.d. Rician cohort, against four deliberate breaks (Table 3, Fig. 6). Every break miscalibrates coverage; the dangerous under-coverage reaches $D^* = 0.515$ under a harder-tissue prior shift and the worst parameter 0.610 under a low-SNR covariate shift. *Weighted / nonexchangeable conformal* [Tibshirani et al., 2019, Barber et al., 2023] cleanly recovers the covariate shifts (SNR worst-parameter $0.610 \to 0.899$, all within 0.011 of nominal; prior D^* $0.515 \to 0.975$) and is honestly limited on the tri-exponential forward-model misspecification: a shift in the conditional law $P(y \mid x)$ is, in general, not repairable by an x-space likelihood ratio (weighting over-corrects D^* to 0.928, leaving a per-parameter spread of 0.028).

The label-free monitor (reusing the deployment-validity concept of a sibling project) **fires on all four observable breaks before coverage fails**: in an SNR-severity sweep, it raises the alarm at SNR 20 (coverage still 0.866) while coverage first crosses the failure line at SNR 13. Yet it is constitutively *blind* to the latent high-D^* gap: on an in-distribution (exchangeable) test the monitor

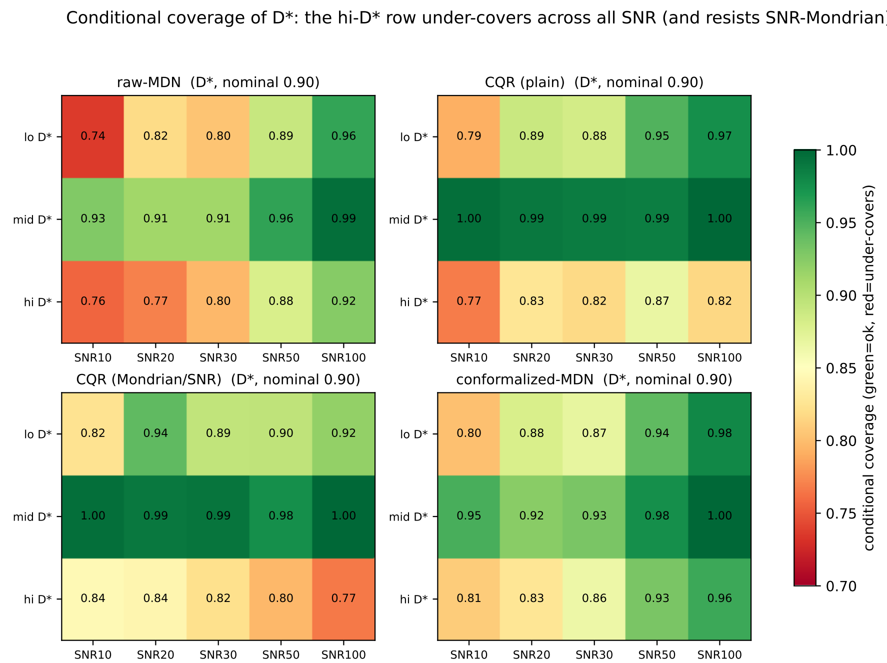

Figure 3: SNR $\times$ D^*-regime conditional coverage. The high-D^* row under-covers across all SNR and resists SNR-Mondrian.

is silent (AUC 0.50) and marginal D^* coverage is nominal (0.900), while the high-D^* conditional coverage is 0.815. Observable shifts are caught; the within-distribution latent-axis failure is not – the same observable-vs-hidden split that recurs across this line of work.

3.5 Acquisition sensitivity: the wall is b-scheme-robust

Does the high-D^* wall move with the b-value scheme? We compare a clinical (11-value), a CRLB-optimal (11-value, greedy D^*-CRLB design), and a dense (22-value) scheme at fixed prior/SNR/seed (Table 4, Fig. 7). The high-D^* marginal coverage barely moves ($0.841 \to 0.844 \to 0.845$) and the resolution ratio CRLB(D^*)/tercile-width stays ≥ 1.05 for *every* scheme. A CRLB-optimal design lowers the absolute CRLB ($1.25 \to 1.05$) but does not remove the wall. Acquisition design *shifts* the wall; it does not erase it – a concrete, honestly negative handoff to acquisition-aware experiment design.

3.6 In-vivo: a qualitative demonstration, no coverage claim

In vivo there are no ground-truth parameters, so realized coverage is *unmeasurable* and the finite-sample guarantee is *not asserted*. We demonstrate the deployed pipeline qualitatively (Fig. 8): the conformalized D^* band widths are wide and heavy-tailed (10/50/90th percentile = $47/64/79 \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$; widest-to-narrowest decile ratio 1.7×), ballooning on voxels where the perfusion compartment is under-identified – the wall, now on signal. Pointed at the deployment data with a synthetic calibration, the deployment monitor **fires** (AUC 0.84): synthetic→deployment is itself a large exchangeability break, and the monitor detects it. That firing is the honest, label-free reason the synthetic guarantee must not be transferred in vivo without re-calibration on matched labeled data – which in vivo does not exist. The monitor turns "no ground truth" into an observable stop

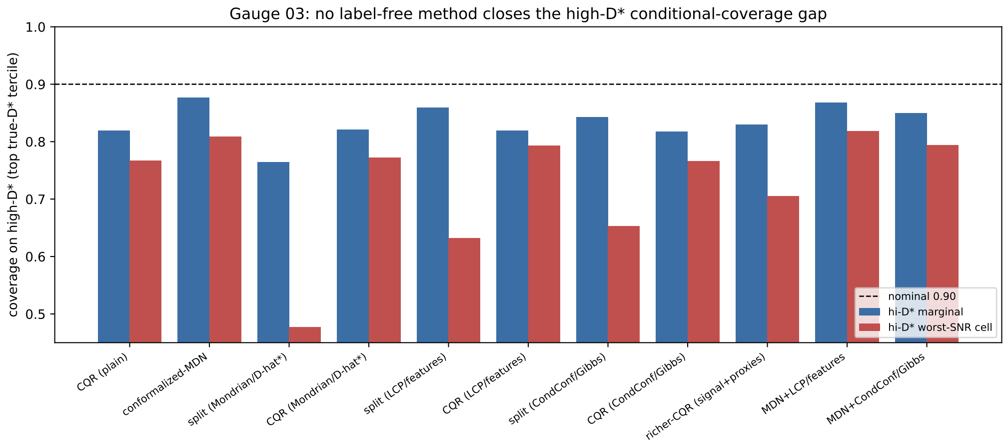

Figure 4: High-D^* coverage (marginal and worst-SNR cell) for eleven label-free conditioning methods; every bar sits below the nominal 0.90.

Table 3: Per-parameter coverage under each break, naive vs. weighted conformal ($\alpha = 0.10$, nominal 0.90), with the deployment monitor's verdict. From `results/robustness_report.txt`.

scenario	method	D	D^*	f	monitor
in-dist (control)	naive	0.909	0.900	0.902	silent (AUC 0.50)
SNR shift (low)	naive	**0.610**	0.744	**0.616**	FIRES (AUC 0.91)
	weighted	0.901	0.899	0.911	
prior shift (harder)	naive	0.970	**0.515**	0.906	FIRES (AUC 0.71)
	weighted	0.900	0.975	0.916	
tri-exp misspec	naive	0.908	0.846	0.910	FIRES (AUC 0.51)
	weighted	0.886	0.928	0.894	
noise-model swap	naive	0.918	0.901	0.904	FIRES (AUC 0.52)

sign. (No usable permissively-licensed, ground-truth-free in-vivo IVIM dataset was available for this study; the demonstration runs on a clearly labeled synthetic stand-in, and dataset choice/licensing and the in-vivo framing are deferred to human sign-off.)

4 Discussion

The thread connecting these results is an *observable-vs-latent* distinction. Conformal prediction guarantees *marginal* coverage and, with weighting or stratification on *observable* quantities (SNR, signal features, estimated density ratios), can recover coverage conditional on those observables. What it cannot label-free-guarantee is coverage conditional on a *latent* axis that the data do not identify. In IVIM that latent axis is the true D^* in its ill-posed regime: the Fisher information there collapses, the CRLB exceeds the bin width, and no observable proxy – including the plug-in $\hat{D}^*$ – recovers the regime. This is why an observable deployment monitor catches every *distribution* shift but is blind to the *within-distribution* high-D^* gap.

The same wall appears in adjacent label-free problems: deployment-validity monitoring is provably blind to shifts that leave observable statistics unchanged (a "hidden channel"), and the absence of in-vivo ground truth (the "Echo tension") forecloses any direct coverage check on real data. We read these not as failures of conformal prediction – whose guarantee is exactly as advertised – but

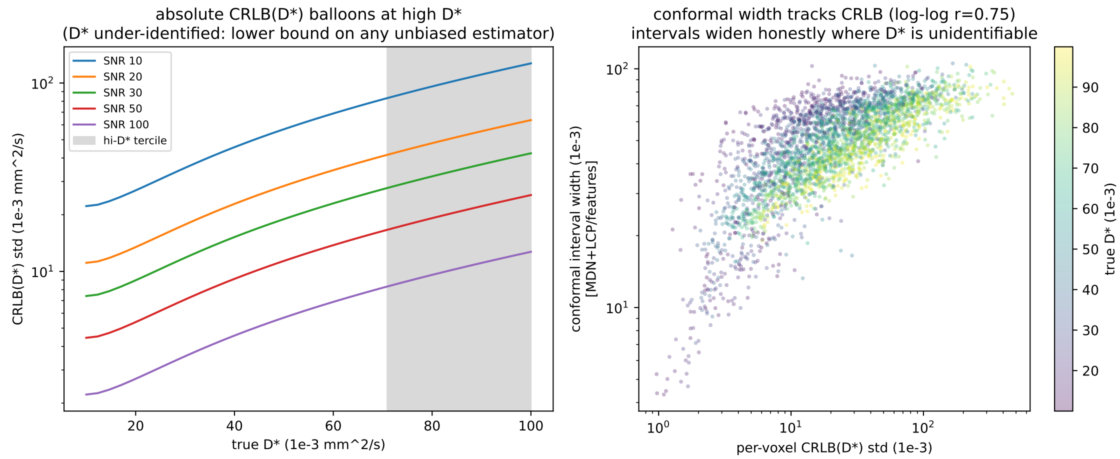

Figure 5: (Left) absolute CRLB(D^*) balloons across the D^* range per SNR (high-D^* tercile shaded). (Right) conformal width vs per-voxel CRLB ($r = 0.75$): the intervals widen honestly where D^* is under-identified.

Table 4: High-D^* coverage and resolution ratio by b-value scheme. From `results/robustness_report.txt`.

scheme	n_b	hi-D^* marg	hi-D^* worst-SNR	CRLB/tercile-width
clinical	11	0.841	0.595	1.25
CRLB-optimal	11	0.844	0.608	**1.05**
dense	22	0.845	0.596	1.06

as a recurring *identifiability* boundary: distribution-free tools deliver everything the data identify and nothing they do not. Practically, this argues for (i) conformalizing a strong model-based base (sharpest valid intervals), (ii) deploying an observable monitor to guard exchangeability, and (iii) reporting the high-D^* compartment as characterized-but-unresolved rather than trusting a point map there.

5 Limitations

Our coverage results are *synthetic-primary* by necessity: conformal calibration needs labels that in-vivo IVIM lacks, so the guarantee is established on a forward model we control, and the in-vivo component is explicitly qualitative with no coverage claim. The synthetic cohort uses uniform priors over published ranges and Rician noise; real tissue is more heterogeneous, and the tri-exponential and prior-shift stresses (Sec. 3.4) are deliberately schematic probes of misspecification rather than a tissue model. The CRLB diagnostic uses the high-SNR Gaussian approximation to Rician noise; the qualitative under-identification of D^* is noise-model-independent (the D^* Jacobian column vanishes as D^* grows), but absolute bound values at SNR $\lesssim 10$ would shift under an exact Rician treatment. Finally, all coverage guarantees assume exchangeability of calibration and deployment data; the monitor quantifies, but cannot by itself repair, departures from it.

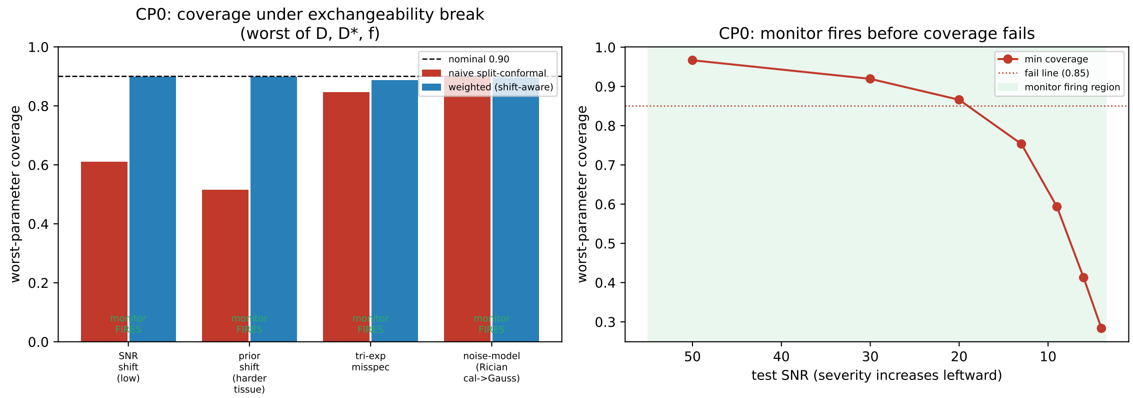

Figure 6: (Left) worst-parameter coverage under each exchangeability break, naive vs. weighted, with monitor status. (Right) SNR-severity sweep: the monitor fires before coverage crosses the failure line.

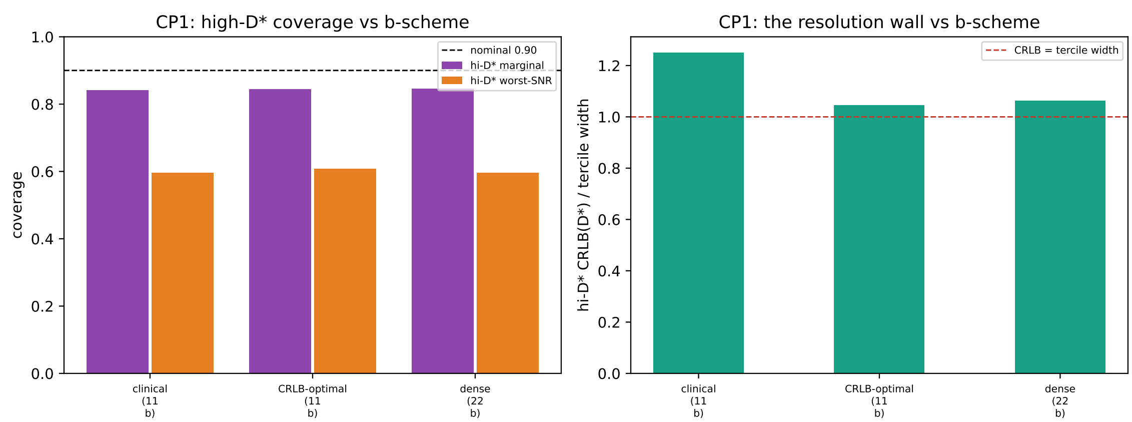

Figure 7: (Left) high-D^* coverage vs. b-scheme. (Right) the resolution ratio CRLB(D^*)/tercile-width stays ≥ 1 for every scheme: the regime is unresolvable regardless of acquisition.

6 Consistency check (reproducibility)

Every numeric claim in this manuscript is drawn from a gated, seeded checkpoint printout across Gauge 01–04. The script `gauge/paper/consistency.py` re-reads the committed report files (`results/*.txt`, `POSITIONING.md`) and asserts that each headline number used here appears verbatim in its source; it prints a one-paragraph pass/fail summary. The full pipeline regenerates deterministically from seed 20260613 (`python -m gauge.baselines`; `gauge.benchmark`; `gauge.conditional`; `gauge.conditional_attack`; `gauge.robustness`; `gauge.invivo`).

7 Conclusion

Gauge brings distribution-free conformal coverage to the ill-posed IVIM inverse problem and benchmarks it against model-based UQ. The naive hypothesis – that the conformal advantage concentrates marginally in D^*/f – is rejected: model-based IVIM UQ is overconfident broadly, conformal restores coverage everywhere, and conformalizing the MDN is the sharpest valid recipe. The gen-

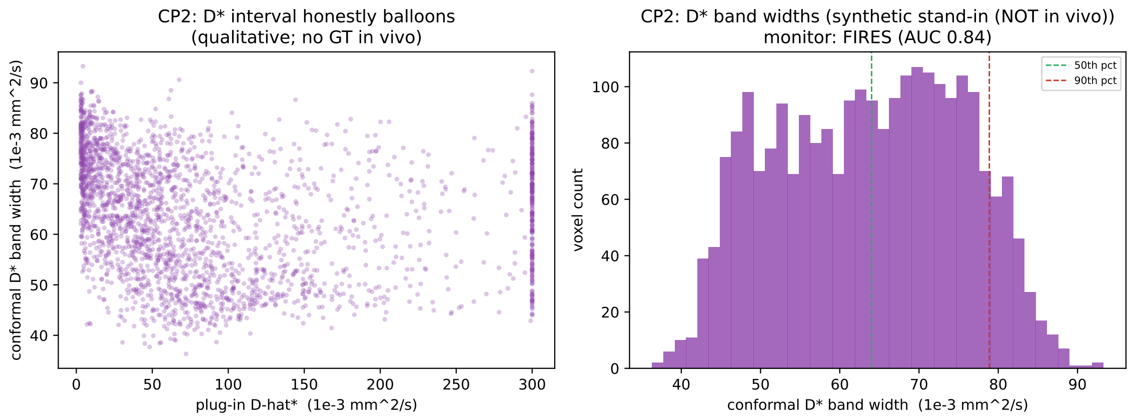

Figure 8: In-vivo qualitative demo (synthetic stand-in). (Left) the conformal D^* band balloons with the plug-in $\hat{D}^*$. (Right) the heavy-tailed D^* band-width distribution; the deployment monitor fires on the transfer.

uine IVIM-specific phenomenon is conditional: the high-D^* regime under-covers universally, and a Fisher-information analysis shows this is an irreducible identifiability limit that no label-free method can close. We therefore characterize the limit – and provide the robustness tooling, acquisition analysis, and qualitative in-vivo demonstration that frame how an honest, distribution-free IVIM uncertainty map should be deployed and monitored.

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